Engineering Mathematics - I · Unit_05 · MATLAB

DRAFT — UE26MA141A · Unit_05 — MATLAB · Complete Unit Master

Notes-first study master · source-cited lecture explanations and compact page takeaways

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the edited lesson notes by topic. Evidence chips point to transcript intervals and slide pages. The complete source transcripts, page text, formula candidates, and image locators are preserved in sibling JSON datasets. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_05 · MATLAB

1 UE26MA141A · Unit_05: MATLAB

This unit master organizes source material for Engineering Mathematics - I, Unit_05, officially titled MATLAB. It includes 6 video transcripts and 13 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Unit map · UE26MA141A · Unit_05: MATLAB

Official catalogue entry: UE26MA141A, Unit_05 (MATLAB). Bucket folder present: yes. Videos: 6; slide decks: 13.

EV0000001

2 Introduction to MATLAB Software

Study focus: Introduction to MATLAB Software. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I'm Sibashankari from the Department of Science and Humanities.

Key details

- For the semester, in the lab session, we will cover basics of MATLAB, programming in MATLAB, plotting 2D and 3D, and then linear algebra, partial differentiation, and solving differential equation.
- And the reference books are "Getting Started with MATLAB" by Rudra Pratap, and the "MATLAB for Engineers" by Holly Moore, and then "MATLAB Programming for

Engineers" by Stephen J.
- The whole syllabus will be covered in 12 sessions, and this is the first session in MATLAB.
- Hope everyone would have downloaded the MATLAB software.
- In case if you have not downloaded, then you can use the installation guidelines link that is attached along with this video.
- As that MATLAB is a paid software, and so if you have institution mail.
- ID, then you can get the access to licensed version.
- Otherwise, you need to download the trial version.
- MATLAB is a software package for numerical computation, visualization, and programming.
- It provides an interactive environment with hundreds of built-in function for technical computation, graphics, and animation.
- The name MATLAB stands for Matrix Laboratory because the basic building block of MATLAB is the matrix.
- Some basic features and commands of MATLAB.
- It consists of MATLAB desktop, a figure window, and an editor window.
- The figure window opens automatically when graphic commands are executed and contains graph created by these command.
- The editor window is used for writing and editing programs.
- This is the place where you can write, edit, create, or save your own program in files called M files.
- We can see the MATLAB desktop, which contains three sub-windows, command window, current folder, and the workspace.
- This is the default layout.
- If you want to change the layout, click on the layout option here and change as per your preference.
- Other two windows are not displayed.
- This is the main window where we write the commands as well as see the outputs.
- In other words, here is the interaction with the software.
- So to type a command, the cursor must be placed next to the command prompt.
- To start with, we can use this command window as a calculator where you can type in commands line by line.

Example or application from the lecture

- To begin with, let us look at the general structure of MATLAB environment.
- Suppose if I select the three column, then the current folder, command window, and the workspace appear in column-wise.
- Suppose if I select it as command window only, then you could see only the command window.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

Once a command is typed, to execute the command, Enter key is pressed.

Key details

- Whenever a calculation is performed, MATLAB will assign the result to the built-in variable ans, unless specified.
- That is the answer variable.
- And ans is the default output variable.
- The easy way to do is, is to use an assignment operator.
- Use the assignment operator, which is equal sign, and then any value or mathematical expression we can give.
- And you should remember that, or make a note of it, the variable must be always on the left, and the expression or the value should be on the right.
- Otherwise, an error will occur if they are reversed.
- And one more thing is that if I type b plus four equal to 66, see, this is an incorrect assignment statement.
- It takes small a and capital A as two different variable.
- That is, the variable should not start with a number.
- It should always start with the alphabet.
- I cannot assign it this way.
- Instead, if I assign a4 equal to 20, it will accept.
- It should start with alphabet.
- And we should avoid using the names of built-in function for variable.
- As my variable, sin equal to 66, then sin variable is created and the value 66 is stored in that sin variable.

Example or application from the lecture

- And after alphabet, a number can be given, any symbol can be given, even underscore can be given, but a space should not be there.
- Suppose if I use sin.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

And how do we know that it is a built-in function?

Key details

- This will give you what are the variables that are there in the workspace.
- Or if you want to know about the size of the data type, then you can just type whos.
- Then it will give you the variable along with the size and the data type.
- And coming to the next one, several commands can be typed in the same line.
- That is, I want to find out the sum of it, then I want to find out the difference, I want to find out a/b, then I want to find out a star b. a star b indicates the multiplication.
- So first this is a+b, and this is a-b, then this is a/b, and this is a*b.

- I'll give it as 33, then, and I don't want to display every time when I give a equal to 33, and when I press the enter key, that value is displayed.
- If I don't want to see that it is being displayed on the screen, what we can do is that for assigning b value, so b equal to, say, if I write 44, I can put one semicolon.
- What is this semicolon will do is that it will suppress the output.
- But in the workspace, if you see, b takes the value 44.
- It is stored there.
- So either what we can do is that you can copy-paste those operation, or you can keep the cursor next to the command prompt and click the up arrow key in your keyboard.
- Then you can see that command.
- You can recall that command.
- So first is addition, subtraction, then I will do multiplication, then I will do the division.
- This is a+b, a-b, a*b, and a/b.

Example or application from the lecture

- Suppose when you are assigning a value to your variable and if you want to check whether you have already used this variable name, or if you want to have the list of variables used, just type who.
- Suppose if you want to modify or make a correction, you can do it.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

We will do some arithmetic operation in MATLAB.

Key details

- Percentage symbol, arithmetic operations.
- Usually, there's no need for commands in the command window, but this is very useful when we are doing programs.
- We will take first problem, and we will execute this.
- Calculate the following using MATLAB command: one by two plus three square, plus four by five into six by seven.
- So actually, we have to do term by term, first is one by two plus three square, and then four by five into six by seven.
- And for multiplication, I should use asterisk symbol six by seven.
- Usually, MATLAB by default displays number with four decimal points.
- If you do so, now if you calculate any problem, say two star sin of 1.4.
- Take on the next problem.
- Find the value of Y equal to E power minus A sine X plus 10 root Y, if A equal to 5, X is equal to 2, and Y is equal to 8.
- So first step, we have to declare these variables.
- A is 5.
- A equal to 5.

- X equal to 2.
- And then Y is equal to 8.
- I need to find out Y is equal to E power minus A sine X plus 10 root Y.

Example or application from the lecture

- The first preference is given to the parenthesis, next is the exponential, then comes the multiplication and division, and finally addition and subtraction.
- Suppose if you see multiplication and division or addition and subtraction on the same line or on the simultaneously if it is present, then it is solved from left to right.
- Suppose if you want to be more accurate, then what you can do is that you can type format long.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 20:00–24:59

plus 10 into square root of Y.

Key details

- Compute sine pi by 4 and 4 into E power 10 sine 30 degree cos inverse pi by 4 and log 10 of exponential E power 2 plus cos pi.
- First, we will do sine pi by 4.
- Don't use the uppercase.
- The next is four into exponential.
- E power 10.
- For exponential, you have to type it as the inbuilt function as exp of 10.
- And if you observe this, the answer is 8.8106 into N power 4.
- So that what it this means.
- Now the next is, we need to find out the value of sine 30 degree.
- You cannot find out for 30 degree.
- You can only find out for the radian.
- that to convert the degree into radian, you have to multiply by pi by 180.
- You need not convert it to radian mode.
- Now coming to the next problem, we need to find out cos inverse of pi by 4.
- This also it is there in the PPT.
- And pi by 4, as I have told you before, pI is the built-in variable by 4.
- And the last problem, log 10 of E power 2 plus cos pi.
- And always that built-in function has to be within the open brackets.

Example or application from the lecture

- Suppose if you want to do only in the degree mode, then you have to use another built-in function, which is S-I-N-D.

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EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

Core explanation · 25:00–29:59

into of E, exponential is exp, 2 plus, cos is also a built-in function, I need to open the bracket, pI.

Key details

- Hope everyone would have got some idea how to work with MATLAB.
- Practice those problems and make yourself familiar with MATLAB interface.
- So we will meet you in the next session.

Example or application from the lecture

- Some more problems are given in the assignment section.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

3 Data types; relational operators; logical operators

Study focus: Data types; relational operators; logical operators. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000026 · EV0000027

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I am Professor Nagesh Ham, Faculty Member, Department of Science and Humanities.

Key details

- This is the second session of MadLab.
- In this session we shall discuss the concept of data types, relational operators and logical operators.
- Data types are those which define the type of data that we are using.
- Every data type stores data that is in a form of a matrix or array.
- Some common data types are integers, floating point numbers, scalar, character, strings and arrays.
- An integer is a whole number.
- There is not a fraction that can be positive, negative or zero.
- Note that when two integers are added, subtracted or multiplied, then the corresponding result is also an integer.
- 2 multiplied by 8, etc When one integer is divided by another integer, the corresponding result may be an integer or a fraction.

- Six by three, that is two.
- Floating point numbers.
- As a name indicates, floating point numbers are numbers that contain floating decimal points.
- Zero point zero one etcetera are floating point numbers.
- When a calculation includes a floating point number, it is called a floating point calculation.
- Any number which is used to represent a quantity is called a scalar.
- This includes integers, complex numbers, floating point numbers.
- character single alpha numeric symbol enclosed in a single code is a character constant.
- The first one is a numerical constant.
- The one which is enclosed in a single code is a character constant.
- Any two or more alpha numeric symbols enclosed in a single code is called a string.
- Arrays List of similar data in a single row or a column is called an array.
- Elements can be numerical or character or strings.
- The four types of arrays are zeros function, i function, once function and ran function.
- Zero's function.
- As the name indicates, it creates a matrix of all zeros.
- It creates an identity matrix.
- One's function.
- It creates a matrix of all ones.
- It generates random numbers between 0 and 1 that are distributed uniformly.

Example or application from the lecture

- Let's start with data types.
- Let's look at this one by one.
- For example, the numbers 10, 0, and minus 25 are integers.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 05:00–09:59

Zero's function.

Key details

- Zero's function, it creates a matrix of all zeros.
- It creates an matrix.
- It creates an identity matrix of order seven.
- Next one's function.
- It creates a matrix of all ones.
- It creates a matrix whose elements are 1.
- It's a matrix of order 10 plus 10.
- Next rank function.
- Rand function is used to generate random numbers between 0 and 1 that are distributed uniformly.

- Here the first number 3 is number of rows and the second number 5 is number of columns.
- This is the corresponding output.
- The numbers are different, but they are between 0 and 1.
- Again use the same command.
- Again you can see that the numbers are different.
- between zero and one that are distributed uniformly using rand function.
- There are six relational operators equal to determine the equality greater than or equal to determine greater than or equal to greater than determine greater than greater than and less than or equal to less than and determined inequally.
- These are the six relational operators we have in MATLAB.
- Relational operator compare the elements in two arrays and return logical true or false values to indicate where the relation holds.
- If the result is true, then it returns the value 1.
- If it is false, then it gives the value 0.
- Remember in MATLAB when we compare two things we use double equal sign.
- 3 is not equal to minus 3.
- Therefore the answer is 0.
- Sixteen star sixty four greater than thousand less nine.
- Look at the answer.
- How the answer is in here.

Example or application from the lecture

- For example zeros let me give some eight is it output it creates a matrix of order eight whose elements are zeros next function.
- Example I sub seven.
- For example, you can give 10.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 10:00–14:59

16 multiplied by 64.

Key details

- It is more than thousand.
- So therefore it gives a value one.
- That one is added with nine.
- Therefore the answer is ten.
- four space minus one space seven five three less than or equal to four.
- Look at the answer.
- It is one zero one because four.
- Less than or equal to four, it is true.
- Therefore the answer is one.
- Minus one less than or equal to four.

- The value is one.
- Seven is not equal to is not less than or equal to four.
- Therefore 0, 0 and 1.
- Sum you can take fourteen then nine three four one four like this equal to four now look at the answer the answer is two here how is it possible.
- It gives the value 0.
- 9 not equal to 4, another 0.
- 3 is equal to 4, another 0.
- There are 3 zeros.
- When I add again I get 0.
- One not equal to four get zero, four equal to four another one.
- It is added because we have used some, therefore the answer is two.
- Is it clear?
- X equals to 10.
- X till a equal to twenty.
- It means see the answer.
- X is ten and X is not equal to twenty.
- It is true the answer is one.
- It is true, therefore it returns the value one.
- Three not equal to three.
- It is false.
- Therefore the answer is zero.
- The logical data type represents true or false states using the numbers 1 and 0 respectively.
- It means we have three logical operators.
- The first one is and second one is r and the last one is negation.
- The logical operator and it means when both operates are true then it gives the value one, otherwise it is zero.
- It is true, that is one when both the operates are true, otherwise the value is 0.
- The result is false.
- Zero one [ASR low confidence; check audio].
- E and B.
- No the answer is zero.
- How it happens?
- One it compares one and zero.
- And one and zero, one it compares element by element, then it gives you corresponding answer.
- It means the logical and it is true when both are true.
- Otherwise the result is false.
- The logical operator R it is true when one or both operands are true.
- You can see when both when one or both operands are true, it is one, otherwise it is zero.
- Therefore got three ones here and one zero.

Example or application from the lecture

- Let us take another example.

- Let us take another example.
- Let us take another example.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

Core explanation · 15:00–19:59

Let us take A equals to one, B is also one, then A B is one, A is zero, and B is one then A one B.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

4 M-Files; Loop Control Statements; Conditional Statements

Study focus: M-Files; Loop Control Statements; Conditional Statements. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000042 · EV0000043

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I am Professor Nagesh Ham, faculty member, Department of Science and Humanities.

Key details

- This is the third session of MATLAB.
- In this session we learn M files.
- That is script files and function files, loop control statements, that is for loops and while loops.
- Finally condition statements.
- In MATLAB we write programs in M files.
- They are called M files because they must have dot M at the end of their name.
- The first one script files and then function files.
- Later we shall see what's the advantage of scries as well as function files.
- A script file is an M file with a set of valid MATLAB commands in it.
- Remember there are two ways to create a script file.
- The first one is this.
- On the command window we just write like this edit and then the file name which we want.
- You can see the output is going to be 600.
- We also find out the product of these three numbers PR O D A star B star C.
- Again, use this button run and you can see the answer is going to be this much.

- Again you give the command whatever you want.
- Let me find out a equal to 10 colon b equal to.
- Sign of y by two sine of pi by two then c equal to twenty right let me find out b equal to a plus b plus.
- Again go to this save.
- Again you can give.
- Let me give EC campus one.
- EC campus one dot m right now.
- the file has been created by the name ecampus1.m.
- You can see the answer is going to be 31.

Example or application from the lecture

- For example it should be like India.m then searching dot m and so on right so they are called m files because they must have dot m at the end of their file name there are two types of m files.
- Let's start with.
- Let us see how to create a script file.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 05:00–09:59

Assume you want to use this after one month of time.

Key details

- You want to compile this.
- Either you can choose the file from here from the current folder or you just type edit then the file name ecampus1.m again you can see the comments on the screen.
- function points A function is a group of statements that together perform a task.
- A function is a group of statements that together perform a task.
- Just clear the screen.
- It let me take such in.
- We run this, you can see answer is going to be 200.
- You have three inputs, A, B, C are the inputs, and the sum and product are the output.
- A B C are the input variables and sum and product are the output variables here.
- Can I cancel this.
- Is it clear?
- is clear then you can see the input variables are a comma b comma c a comma b comma c like this now remember the file name should be same as the one which we have used for creating the script file clear so this should be same as such in without M we should not use dot M here.
- Is it clear?
- Then we can use sum equal to A plus B plus C.
- Then Product PROD equal to A star B star C, B star C.

- Remember before we run this we have to say this.
- Remember again the file m should be same as the one what we have used for creating script file.
- We have saved a function file by searching.m.
- Remember in order to compile this we should use command window.
- search in of you can use whatever variables you want you can give okay I gave one comma one comma hundred right okay this is equal you can write it as s for sum p for product.

Example or application from the lecture

- So let us take an example.
- such in dot m let's create a script file for this searching now you give the command e equal to one B equal to two C equal to hundred then find out the sum equal to A plus B plus C then run this against answer is 103 then find out your product equal to a star b star c.
- So for that, what you need to do, you just clear the screen and remember for this particular example.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 10:00–14:59

Equal to like this.

Key details

- You can see S is 102, the sum is 102, and the product is 100.
- See if you want to.
- Add some of the three numbers.
- What we should do?
- We have to manually update a script file every time.
- Just copy paste here.
- You can change the number here.
- Then hundred, then thousand.
- Use we can see the output.
- It means instead of manually updating the script each time, we can make program more flexible by converting it to a function.
- Is it clear?
- Clear the screen.
- Next we shall move on to low control statements.
- Using loop control statements, we can repeatedly execute a block of code.
- There are two types of loops.
- One is for loops and while loops.
- For loop is used to repeat a statement or a group of statements for a specific number of times.
- For loop is used to repeat a statement or a group of statements for a specific number of times.

- Remember here what we see DISP of A, this is not a display command.
- It displays the values of A without a variable name.
- Is it clear?
- You can see the this is A is a variable whose values are 1, 2, 3, 4.
- Displays the values of A without printing a variable name that is A.
- Here m start from 1 goes up to 3 so you can write number equal to 1 divided by m plus 1 then n.
- When m is two number is one zero point double three double three and so on.
- Is it clear?
- Is it clear?
- While loop repeatedly execute program statements as long as the expression remains true.
- Then now the while loop starts.
- And the condition here is A is less than fifteen.

Example or application from the lecture

- Now for example, for the same command, we can give some number like this.
- Let's talk about for loops.
- For example, we consider this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 15:00–19:59

the while will run till the value of A is less than 15.

Key details

- Next we print the value F P R I N D print f printf value of e colon percentage D slash N.
- It means it prints a current value of A and then A equal to A plus 1 is executed.
- It means this is the output for the following command.
- It means we first define a equals to 10.
- We define the variable a equal to 10.
- Now the while loop starts and the condition for a is a is less than 15.
- It means that the while loop will run till the value of A is less than 15.
- Currently the value of A is 10.
- The line A equal to A plus 1 add 1 to A and then while conduction is checked.
- That's why the output is like this 10 to 14.
- And these are the some of the examples what we have discussed.
- There will be some situation where a program or a particular block has to be executed only when a specific condition is true.
- The condition statements will be useful in such situations.
- The following are the condition statements that we can use in MATLAB.
- If and if else and if else if else if else and switch case.
- So these are the some of the condition statements what we use in MATLAB.
- If and statement.

- Number equal to 25.
- Here you can see that number is 25.
- So this 25 is more than 20, so it is true, therefore it prints the output like this.
- Because the condition is true here.
- Number is 25.
- Here the condition is false, that's why it displays nothing on the screen.
- So next we have if else and statement.
- An if and statement can be followed by an optional L statement which executes when the expression is false.
- If the number is less than 20, so this is false, therefore it prints the second line.
- The number is not less than 20.
- The number is not less than 20.
- So if this condition is not true, therefore it executes the next line.
- Next if else if else if else and so this is a chain.
- So next we have a switch statement.
- The switch statement is used to test for equality against a set of known values.

Example or application from the lecture

- After understanding loop control statements, so let's move on to conditional statements.
- Let us model the loop a condition statements.
- It decides whether a particular block of code has to be executed or not based on the given Boolean condition that is 0 or 1.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

Core explanation · 20:00–24:59

This one when we run the file it displays the following, is it?

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

5 Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling

Study focus: Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000062 · EV0000063

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I am Professor Nagesh Ham, faculty member Department of Science and Humanities.

Key details

- This is the fourth session of MATLAB.
- In this session we learn plotting using MATLAB.
- There are three cases.
- Case number one, Cartesian curves that is y equal to f of x.
- Case number two polar curves that is R equal to F of theta.
- Case number three parametric curves x equal to f of t and y equal to g of t.
- To plot the graph of a function, we need to take the following steps.
- We first define the variable x, then specify the range of values for the variable x.
- We define the function y equal to f of x.
- Finally we use plot command in order to plot the Cartesian curve that is y equal to f of x.
- The plot command is used to create two dimensional plots.
- Before we plot this graph, we have to use the first one.
- Then as you can see, we have to have a 1.
- this is a three crossary matrix.
- And this is a normal or usual matrix multiplication what we know A star b.
- But if you look at A dot star B.
- As you can see the first element of this A is multiplied with this 4.
- Then similarly, multiplication of 2 and 5, so the corresponding number is 10.
- This is how the dot operator works.
- The dot operator is used with multiplication, division and exponentiation to create element wise operation.
- That's what we see here.
- If you look at A star b this is a usual matrix multiplication what we know, right?
- I repeat, the dot operator is used with multiplication, division, exponentiation to create element wise operation.
- And we know this is a procedure.
- We first define the variable x and then we specify the range of values for the variable x.
- Then we define the function y equal to f of x for which we need to plot the graph.
- Then finally we use plot of x comma y command.
- Let x equals two so minus 100.
- This is colon.
- It means x starts with minus 2, it goes up to 4 with a step size 0.01.
- Then define the function y equal to.

Example or application from the lecture

- Let's start with the Cartesian case that is y equal to f of x.
- In order to understand this concept, let us look at the example y equal to 3.5 power 0.5 x into cos of 6x where x lies between minus 2 and plus 4.
- Let's understand the concept of dot operator in Matlab.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 05:00–09:59

three point five dot this is a dot operator remember cap minus zero point five x again this has to be star x then dot operator star cos of six times x right then finally use the command plot of x comma y.

Key details

- What happens if you give plot of y comma x.
- This is shape of the graph.
- Way come my X.
- This is the shape of the curve.
- Is it clear?
- This is the error what we get.
- Is it clear?
- y equal to 3.5 powers minus 0.5 x into cos of 6x.
- Square then plot of X comma y.
- Then plot of x comma y.
- When we decrease the increment, the graph becomes smoother.
- When we decrease the increment, the graph becomes smoother.
- That's it.
- Is it clear?
- We know that.
- A function of two independent random variables defines a surface in three dimensional space.
- Step number one.

Example or application from the lecture

- For example, let me cancel this, let me delete this, right?
- So let us take another example y equal to f of x square.
- So in this example we will draw two graphs with the same function, but in the second time we will reduce the value of increment.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

Core explanation · 10:00–14:59

We first create a set of points over the domain of the function using mesh grid command.

Key details

- Next we assign the function itself.
- Finally we use a command called surf.
- to plot the surface.

- clear the screen we can write it is x y you can also give capital x y there mesh grid mesh grid of minus two right colon minus point two that is the step size then colon you can give two.
- You start from minus two it goes up to four right then you c call the command z equal to the function is x into x dot product exp for exponential minus x square minus x square means dot cap square minus y dot cap square.
- Finally we use a command surf of x comma y comma z right.
- Is it clear?
- A function of two independent random variables defines a surface in three-dimensional space.
- You can rotate however you want.
- Is it clear?
- See this how you can rotate.
- Is it clear?
- Using MATLAB we can add title labels along the x axis as well as y axis of the graph.
- Commands generate labels along X axis as well as Y axis.
- Finally the title command allows you to put a title on the graph.
- Y label you can say s single code sign of X right then you can give title so title it is.
- Sine of X graph.
- Is it clear?
- And then you can see the corresponding answer is going to be the corresponding graph is this.
- As you can see X axis labeled by X.

Example or application from the lecture

- So in order to understand this concept let us take a simple example let us plot the surface for z equal to x into e power minus of x square plus y square.
- Said let's look at the illustration.
- Okay, let us take a simple example another example you can take sine x cos y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 15:00–19:59

Y axis is labeled by sine of x and this is the name of the graph.

Key details

- Sine of x graph.
- Is it clear?
- And this is the corresponding answer.
- 0 starting from 0, increment value 0.01 colon 2 pi 2 star pi right.
- Then define the function by rho.
- Rho equal to we are looking for sine 2 theta sine of 2 star theta right multiplication dot operator should be there then cost of two star theta is it clear then use polar plot command polar plot it is simply theta comma row is clear so this theta gamma row.
- Is it clear?

- Then polar plot.
- And this is a graph.
- Cas2 star theta here.
- And then copy paste it here.
- And this is a graph of cos2 theta looks like.
- Is it clear?
- Sims function creates a variable dynamically.
- Is it clear?
- In order to plot parametric curve okay before that what is the equation of a parametric curve?

Example or application from the lecture

- the equation of a polar curve is given by r equal to f of theta clear so in order to plot a polar curve we use this command polar plot of theta comma rho here theta is the angle in radians and rho is a radius value for each point.
- Let us take a simple example r equal to sine 2 theta into cos 2 theta right.
- Let's look at another example, let us take you can take r equal to one minus cos theta, you can take some example.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 20:00–24:59

Is it clear?

Key details

- In order to plot this we use this simple command.
- Then we use x t equal to sine t means sin of t.
- colon then y t equal to cos of t then Z t equal to T right so finally we use F plot three X T Y T Z T clear So this is the shape of the graph what we get.
- Is it clear?
- You just ignore these three.
- You get an error.
- Is it clear?
- Therefore, we need to use f plot 3 x t y t and z t.
- parametric curve x equal to e power minus t by 10 sin phi t y equal to e power minus t by 10 cos phi t z equal to t over the interval minus y and plus minus 10 to plus 10.
- Is it clear?

Example or application from the lecture

- So it is other than minus y to plus phi we use another command which we shall see in the next example.
- Take an example let us plot a parametric curve x equal to sin t y equal to cos t z equal to t over the default parameter minus y2 plus y.

- Suppose intentionally we make a mistake.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

6 Video

Study focus: Video. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000082 · EV0000083

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this MATLAB session, we explore partial derivatives using MATLAB.

Key details

- And then after that we need to define the function.
- And then the partial derivative of the function f with respect to x is written as dif.
- of f comma x and with respect to y we write it as dif f of f comma y.
- And then the function f is equal to sine of x.
- plus y cube, we're write it as y cap three plus x power ten that is x cap ten minus y square plus log of x.
- If f is equal to x square plus 2y square minus 22, we need to find fxx and f y.
- The symbolic variables x and y are defined and f is equal to the function x square plus 2y square minus 22 is defined and then dif f of f x2 is nothing but partial derivative of f with respect to x and again also with respect to x.
- If f is equal to x y q plus tan x plus cos of root log x then find dfx here the.
- If f is equal to xy cube divided by x plus y, then find fx f y fx f x y and f yx.
- We know that the code to find the partial derivative of.
- And dif of f comma x comma y is nothing but partial derivative of f with respect to x first and then with respect to y.
- And at last dif of f x is nothing but partial derivative of f with respect to y first and then with respect to x.

Example or application from the lecture

- If f is equal to sine of x plus y cube plus x to the power 10 minus y square plus log of x as a given function, then find fx and f y.
- Here we need to find the partial derivative of the given function f with respect to x and with respect to y also.
- We'll move to the next example.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 05:00–09:59

If u is equal to x square y plus y square z plus z square x, we need to find the first order partial derivative of u.

Key details

- Dif of u x.
- If u is equal to root of x square plus y square plus z square plus tan inverse of x y, find the first order partial derivative of u.
- We are defining.
- It is nothing but tan inverse is written as a tan of x into y.
- A partial derivative first order partial derivative of u with respect to x y z r also written as d ff of u x1 d ff of u y one df of u z1 respectively.
- And this is the output we are obtaining for this problem.
- When all the second order second partial derivatives of f of x y is equal to log of one by x minus one by y at one comma two.
- Partial second order second partial derivatives.
- And here we are initially defining.
- A of x y as dif of f of x2, partial derivative of f with respect to x twice, and partial derivative of f with respect to y twice as b of x comma y.
- And C of x comma y is nothing but partial derivative of f with respect to x first and then with respect to y.
- D of x y is nothing but partial derivative of uf with respect to y, comma.
- X is nothing but derivative of f with respect to y first and then with respect to x.
- And also we are continuing the same code with a of 1 comma 2, b of 1 comma 2, and c of 1 comma 2 d of 1 comma 2 because at upper point 1 comma 2 we need to find the second partial derivatives of the function.
- And this is the output we obtained.
- And at 1 comma 2 we are obtaining 0 minus 3 by 4, 1 and 1.
- If u is equal to e power x y plus y z plus zx, then find u x y uyz and uz.
- Dif of u y z is nothing but partial derivative of u with respect to y first and then with respect to z.
- DIFF of U Zx is nothing but partial derivative of U with respect to Z first and then with respect to X.
- If f is equal to e power x y z then find f x z f y z so symbolic variables x y z are defined and also the even function is defined in the second step.
- So d f of x f x y z is nothing but partial derivative of f with respect to x first and then with respect to y and at the last with respect to z.
- In the similar way, f xs z is nothing but partial derivative of f with respect to x first and again also with respect to x and at last with respect to z.
- The f of f y z is nothing but partial derivative of uf with respect to.
- Y first and again with respect to y and at last with respect to z.

- Next, we'll move on to the Telia series expansion of the function of single variable.

Example or application from the lecture

- So we need to define all the three symbolic variables s y s xy z and then the given function is defined.
- one is nothing but partial derivative of the given function u with respect to x and u y and u z is also done in a similar way and this is the output we are obtaining.
- If z is equal to sin two x plus three y as a given function, we need to find the first order partial derivative of z.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

Core explanation · 10:00–14:59

We need to define the symbolic variables in the initial stage and also the function has to be defined.

Key details

- T is equal to tailor of f of f, comma.
- Expansion point to order 3.
- Expand the function about the point x is equal to 2.
- comma with above the point X is equal to 2, so we are defining it as expansion point.
- Two expansion point two.
- And with order.
- [ASR low confidence; check audio].
- Next function f of x is equal to log of cos x about the point x is equal to pi by 3 up to 5th degree terms.
- It has also worked in a similar way.
- Here the expansion point is.
- Pi by 3 and order it is up to 5th degree terms, so we are taking order 6.
- Expand f of x is equal to log of secant x about the origin up to six degree terms.
- S y must x and also after that the function log of secant x is defined.
- T is equal to tailor of f with the order seven.
- Here it is about the origin, so no need to mention the expansion point.
- It is up to six degree terms, so we are writing the order seven.
- f of x is equal to sine of log of x square plus 2x plus 1 about the origin up to 6 degree terms.
- So same way x is defined and the function f is defined and also it is also about the origin.
- Is the code we are writing.
- That is about the origin.
- expand f of x comma y is equal to e power x into cos y above the point x is equal to 1 and y is equal to pi by 4 up to three degree terms.
- And the function is f is equal to exponent of x.
- into cos y.
- Taylor of the function yf.

- One comma [ASR low confidence; check audio].
- I buy sport.

Example or application from the lecture

- So the symbolic variable x is defined and the given function log of cos x is defined.
- The same way t is equal to tailor of the given function f.
- It is about the point XY that given it as.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 15:00–19:59

I have given up to three degrees terms.

Key details

- So next problem is f of x y is equal to x cube plus y cube plus xy square about x is equal to 1 and y is equal to 2 up to 4th degree terms.
- It is also the similar problem to which we just now.
- Next function e power y into log of 1 plus x above the original to fourth degree terms.
- comma x y so it is about the origin so no need to mention the points over there and it is up to four degree terms so the order five and this is the result we are obtaining f of x y is equal to d power x into tan y above the origin up to fifth degree terms.
- That is about the origin no need to mention the points over there in the line in the statement and it is up to fifth degree terms so the order six and this is the output which we are obtaining.

Example or application from the lecture

- For this given problem.
- The function is also defined and here the points that are given at as 1 comma 2 and it has up to 4 degree terms.
- So symbolic variables are defined and also the given function is defined.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

7 Video

Study focus: Video. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000098 · EV0000099

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session we are going to learn about higher order differential equations in MATLAB.

Key details

- Here our first question to solve the higher order differential equations that is y double dash plus phi y which is equals to zero.
- Things of y of x which is nothing but we are.
- Defining the variables I mean symbolic variables next ODE which is equals to diff of y comma x comma 2 plus phi star fi double is equals to zero.
- Here O D is nothing but ordinary differential equations and d if f is nothing but differentiation and d if f of y comma x comma two is nothing but second derivy q plus phi into y which is double is equal to zero.
- Where double is equals to which means defining an equation symbolically that is left hand side which is equals to right hand side.
- Next to for the solution we are taking y solve of x which is equals to d solve of ODE.
- Where d solve of.
- Where D solve which is equal to differential equation solver of ODE function.
- First we are defining the symbols that is y of x.
- Next, we are defining the question that ODE which is equals to differentiation of y, comma x, comma 2.
- 5 star y which is double is equals to 0.
- Semicolon next y solution of x which is equals to D sol of Ode.
- Moving on to the similar type of the question that is y double dash plus seven y which is equals to zero and we'll get the solution that is y solution of x which is equals to c1 cos root seven x minus c2 sin root seven x.
- This is of similar portion instead of 7 I have taken 18y which is equal to 0.
- Next moving to the higher order higher order differential equation.
- That is 12y double dash plus 10y dash minus 8y which is equals to 0.
- Here again we are defining the variables that is of since of y of x and od which is equal to 12 star.
- The sense 12 into differentiation this is of second derivative plus 10 into first derivative minus 8y which is double is equals to 0 and y solve of x is equals to d solve of od.
- Next moving on to the matlab I'm taking a section break.
- This is of defining the symbols Simps of Y of X.
- Next, defining the question.
- Defining the question as ODE which is equal to 12 star second derivative d i f of y comma x comma to plus 10 star differentiation of y, comma x comma 1, which is the first derivative minus 8 star y.
- And double is equal to 0.
- I am closing with semicolon.
- Next moving on to the Y solution of X, which is equals to D solve of ODE.
- x by 2 plus even e power minus 4x by 3.

- Next moving on to the next question of similar question that is 8y double dash plus 2y dash minus y is equal to 0 as of same question.
- Next this is of order 4.
- Hereof also seen we are defining the symbols that is of seams of y of x and od is equals to differentiation of fourth order of.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

Core explanation · 05:00–09:59

Fourth the different derivative differentiation of y, comma four plus eight into second.

Key details

- Second derivative plus 16y which is equal to 0 and y solve of x is equals to d solve of ODE.
- Here of order 3 of similar question.
- Next here I have taken d cube minus 3 d square plus 4.
- into x which is equal to 0.
- Hereof also same we are defining the symbolic variables that is y of x and o d e which is equals to third order derivative minus 3 into second derivative.
- Plus 4y which is equals to double is equals to 0 and y solution which is equals to desolve of y.
- Here I am defining the symbols.
- This is of y of x.
- Next defining the ODE function ODE which is equal to DIFF of differentiation of Y, X, comma, 3.
- Minus 3 star d if of differentiation of y, x, 2.
- Plus 4 star y double is equals to 0.
- I am closing with semicolon and y solve.
- Y solve of x which is equals to D solve of Ode.
- I'm getting output as C1 E power 2X plus C3 E power minus X plus C2 X E power 2X.
- Next, moving on to the next question.
- Here thereof initial conditions.
- First here defining as the symbols of variables.
- Sims of y of x and we are defining the question as ODE which is equals to second derivative.
- Minus 5 into first derivative plus 6y which is double is equals to 0 and dy which is equal to first derivative we are defining and d2y which is equal to second derivative we are defining.
- And C O N D 1, which is of condition 1, which is equal to y of 0, double is equals to minus 1, and condition 2, which is equals to dy of 0, double is equals to 0.
- And y solution instead.
- We are taking all the three conditions that is of D solve of ODE, condition one, comma condition two.
- Here Dispov Y solve, which is nothing but we are displaying the.
- and EZ of EZ plot of Y sort which is nothing but easy to plot the graph.
- So first we are defining the scenes since of x.

- Next define the question as ODE which is equal to DIFF of y comma x comma two minus 5 into PIF of y, comma x.
- This is a this is our first derivative plus 6 star y which is double is equals to 0.
- I am again closing with semicolon and.
- First derivative which is equal to dif of y, comma x and d2 y which is equal to second derivative.
- Dif of y, comma x, comma two and I am defining the conditions.
- It is of condition one which is equal to dy which is equals to or y of 0.
- Y of 0 which is equals to minus 1.

Example or application from the lecture

- Let us see in MATLAB code.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

Core explanation · 10:00–14:59

Dou is equals to minus one.

Key details

- So second condition two which is equal to D2y.
- Dy of zero, which double is equals to zero.
- I'm closing with semicolon.
- Next Y solve Y solution of X.
- which is equal to D solve of ODE, comma, condition one, comma, condition two.
- I am taking all the three conditions.
- and closing with semicolon and I am displaying the solution display of y solution so I am getting the output that is 2 e power 3x minus 3 e power 2x and for the plot I am using.
- E Z plot that is of easy to plot the Y solution.
- Next moving on to the similar questions.
- And the question is y triple dash minus 6y double dash plus 1 y dash minus 6y is equals to 0.
- For this question also, we are going to plot the graph.
- You are getting this type of graph.
- Next moving on to the single question.
- That is of conditions with y of 1 is equals to 0 and y dash of 1 is equals to minus 1.
- Next question, next conditions you are defining.
- Next solution, you are displaying the solution or you are simplify the solution.
- Next moving on to the a z plot of solution.
- Here I have taken one step extra that is why solution is equals to simplify of y solve.
- This is nothing but we are getting the output and we are solving again the same solution and we are displaying the solution.
- We are plotting for the graph for the same question.
- This type of answer we are going to get that is minus e power 2 minus 2x of x minus 1.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

8 Source slide deck: Slides

Study focus: Source slide deck: Slides. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000384 · EV0000385

SHORT-NOTE RECAP

Study notes · Source slide deck: Slides

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MATHEMATICS-I

- Department of Science and Humanities

Slide 2 · MATLAB – Plotting

- ➢To plot the graph of a function, we need to take the following
- ➢Following example would demonstrate the concept.
- ➢Let us plot the function $y = 3.5 - 0.5x\cos 6x$ for $-2 \le x \le 4$.

Slide 3 · MATLAB – Plotting, Continued...

- ➢The script file is as follows:
- >> x=[-2:0.01:4]; (Here 0.01 is the spacing value)
- >> y=3.5.^(-0.5*x).*cos(6*x);

Slide 4 · MATLAB – Plotting, Continued...

- ➢When we run the file, MATLAB displays the following plot:

Slide 5 · MATLAB – Plotting, Continued...

- ➢Let us take another example to plot the function $y = x2$.
- ➢In this example, we will draw two graphs with the same function, but
- in second time, we will reduce the value of increment.

Slide 6 · MATLAB – Plotting, Continued...

- ➢When we run the file, MATLAB displays the following plot:

Slide 7 · MATLAB – Plotting, Continued...

- Change the code file a little, reduce the increment to 5.
- >> plot(x, y). When you run the file, MATLAB displays the following

Slide 8 · MATLAB - Three Dimensional Plots

- ➢Three-dimensional plots display a surface defined by a
- ➢To define the function $g = f(x, y)$, we first create a set of
- Finally, we use the surf command to create a surface plot.

Slide 9 · MATLAB - Three Dimensional Plots, Continued...

- ➢The following example demonstrates the concept:
- ➢Let us create a 3D surface map for the function $z = xe - x2 + y2$
- ➢The script file is as follows:

Slide 10 · MATLAB - Three Dimensional Plots, Continued...

- ➢When we run the file, MATLAB displays the following 3-D map:

Slide 11 · MATLAB - Three Dimensional Plots, Continued...

- ➢Let us consider another example to create a 3D surface map for
- ➢The script file is as follows:
- >> z=sin(x)+cos(y);

Slide 12 · MATLAB - Three Dimensional Plots, Continued...

- ➢When we run the file, MATLAB displays the following 3-D map:

Slide 13 · MATLAB - Three Dimensional Plots, Continued...

- ➢Let us consider another example to create a 3D surface map for
- ➢The script file is as follows:
- >> [x,y] = meshgrid(-2:.2:2);

Slide 14 · MATLAB - Three Dimensional Plots, Continued...

- ➢When we run the file, MATLAB displays the following 3-D map:

Slide 15 · Adding Title and Labels on the Graph

- ➢Using MATLAB we can add title, labels along the x-axis and
- ➢The xlabel and ylabel commands generate labels along x-axis
- ➢The title command allows you to put a title on the graph.

Slide 16 · Adding Title and Labels on the Graph

- ➢When we run the file, MATLAB displays the following plot:

Slide 17 · MATLAB – Polar Plots

- ➢To plot a curve in polar coordinates, we use the following
- ➢polarplot(theta,rho). Here theta is the angle in radians and rho is
- ➢Following example would demonstrate the concept.

Slide 18 · MATLAB – Polar Plots, Continued...

- ➢The script file is as follows:
- >> theta = 0:0.01:2*pi;
- >> rho = sin(2*theta).*cos(2*theta);

Slide 19 · MATLAB – Polar Plots, Continued...

- ➢When we run the file, MATLAB displays the following plot:

Slide 20 · MATLAB – Polar Plots, Continued...

- ➢The script file is as follows:
- >> theta = 0:0.01:2*pi;

Slide 21 · MATLAB – Polar Plots, Continued...

- ➢When we run the file, MATLAB displays the following plot:

Slide 22 · MATLAB – Polar Plots, Continued...

- ➢The script file is as follows:
- >> theta = 0:0.01:2*pi;

Slide 23 · MATLAB – Polar Plots, Continued...

- ➢When we run the file, MATLAB displays the following plot:

Slide 24 · MATLAB – Parametric Plots

- ➢fplot3(xt,yt,zt), which plots the parametric curve xt=x(t), yt=y(t),
- ➢Following example would demonstrate the concept.
- ➢Let us plot a parametric curve: x=sin(t); y=cos(t); z=t over the

Slide 25 · MATLAB – Parametric Plots, Continued...

- ➢The script file as follows:

Slide 26 · MATLAB – Parametric Plots, Continued...

- ➢When we run the file, MATLAB displays the following plot:

Slide 27 · MATLAB – Polar Plots, Continued...

- 10 ➢Plot the parametric curve $x = e-t \sin 5t$; $y =$
- Note that fplot3(xt,yt,zt, [tmin tmax]) plots xt = x(t), yt = y(t),
- and zt = z(t) over the interval tmin < t < tmax.

Slide 28 · MATLAB – Parametric Plots, Continued...

- ➢The script file as follows:
- >> xt = exp(-t/10).*sin(5*t);
- >> yt = exp(-t/10).*cos(5*t);

Slide 29 · MATLAB – Parametric Plots, Continued...

- ➢When we run the file, MATLAB displays the following plot:

Slide 30 · Slide page 30

- no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection.

EV0000384 · EV0000385 · EV0000386 · +85 source units in pack.json

9 Source slide deck: UNIT 1

Study focus: Source slide deck: UNIT 1. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000562 · EV0000563

SHORT-NOTE RECAP

Study notes · Source slide deck: UNIT 1

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · [Image text, OCR]

- Department of Science and Humanities

Slide 2 · Partial Differentiation – level 1

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · Partial Differentiation – level 1

- disp('Partial derivative with respect to x:')
- disp('Partial derivative with respect to y:')
- Partial derivative with respect to x:

Slide 4 · Partial Differentiation – level 1

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 5 · Partial Differentiation – level 1

- disp('Partial derivative w.r.t y:')
- Partial derivative w.r.t x:
- Partial derivative w.r.t y:

Slide 6 · Partial Differentiation – level 1

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 7 · Partial Differentiation – level 1

- disp('For z2= exp(x*y):')
- disp('Partial derivative w.r.t x:')
- disp('Partial derivative w.r.t y:')

Slide 8 · Partial Differentiation – level 1

- For z2 = exp(x*y):
- Partial derivative w.r.t x:
- Partial derivative w.r.t y:

Slide 9 · Partial Differentiation – level 1

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 10 · Partial Differentiation – level 1

- dz_dy = diff(z, y); % Second derivatives
- d2z_dx2 = diff(z, x, 2);
- d2z_dxdy = diff(diff(z, x), y); % Display results

Slide 11 · Partial Differentiation – level 1

- disp('Second derivative w.r.t x:');
- disp(d2z_dx2); disp('Second derivative w.r.t y:');

Slide 12 · Partial Differentiation – level 1

- fprintf('Second order Partial derivative wrt x =%s\n',char(d2z_dx2))
- fprintf('Second order Partial derivative wrt y=%s\n',char(d2z_dy2))
- fprintf('Mixed Partial derivative =%s\n',char(d2z_dxdy))

Slide 13 · Partial Differentiation – level 1

- Second derivative w.r.t x:
- 6*x*y Second derivative w.r.t y:
- 6*x*y Mixed derivative:

Slide 14 · Partial Differentiation – level 1

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 15 · Partial Differentiation – level 1

- d2z_dxdy = diff(dz_dx, y);
- d2z_dydx = diff(dz_dy, x); % Display results
- disp(d2z_dxdy); disp('d2z/dydx = ');

Slide 16 · Partial Differentiation – level 1

- d2z/dxdy = 4*x*y d2z/dydx =

Slide 17 · Partial Differentiation – level 1

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 18 · Partial Differentiation – level 1

- y^2*sin(x*y)
- x^2*sin(x*y)
- cos(x*y) - x*y*sin(x*y)

Slide 19 · Partial Differentiation – level 1

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 20 · Partial Differentiation – level 1

- dz_dr = simplify(diff(z,r));
- dz_dtheta = simplify(diff(z,theta));

Slide 21 · Partial Differentiation – level 1

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 22 · Partial Differentiation – level 1

- syms x y r theta
- dz_dx = diff(z,x);
- dz_dy = diff(z,y);

Slide 23 · Partial Differentiation – level 1

- yr = r*sin(theta); dx_dr = diff(xr,r);
- Chain = subs(dz_dx,[x y],[xr yr])*dx_dr + ...
- Direct = diff(subs(z,[x y],[xr yr]),r);

Slide 24 · Partial Differentiation – level 1

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 25 · Partial Differentiation – level 1

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 26 · Partial Differentiation – level 1

- y(2x+y-e*y)

Slide 27 · Partial Differentiation – level 1

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 28 · Partial Differentiation – level 1

- dz_dy = diff(z, y); % Required expression
- result = x*dz_dx + y*dz_dy; % Simplify result
- disp('Value of x*dz/dx + y*dz/dy = ');

Slide 29 · Partial Differentiation – level 1

- Value of x*dz/dx + y*dz/dy = 0

Slide 30 · Partial Differentiation – level 1

- dz_dx = diff(z, x);
- dz_dy = diff(z, y);
- LHS = x*dz_dx + y*dz_dy;

Slide 31 · Partial Differentiation – level 1

- 2*x^2 + 2*y^2
- 2*x^2 + 2*y^2

Slide 32 · Partial Differentiation – level 1

- dz_dx = diff(z, x);
- dz_dy = diff(z, y);
- LHS = x*dz_dx + y*dz_dy;

Slide 33 · Partial Differentiation – level 1

- LHS =2*x^3/y
- RHS =2*x^3/y

Slide 34 · Partial Differentiation – level 1

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 35 · Partial Differentiation – level 1

- Maclaurin Series: 1 + x - x^3/3

Slide 36 · Partial Differentiation – level 1

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 37 · Partial Differentiation – level 1

- 1 + x + y + x^2/2 + y^2/2 + x*y

Slide 38 · [Image text, OCR]

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000562 · EV0000563 · EV0000564 · +111 source units in pack.json

10 Source slide deck: UNIT 1 CONTINUED

Study focus: Source slide deck: UNIT 1 CONTINUED. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000676 · EV0000677

SHORT-NOTE RECAP

Study notes · Source slide deck: UNIT 1 CONTINUED

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · [Image text, OCR]

- Department of Science and Humanities

Slide 2 · Partial Derivative-Class II

- T = 100 - 2*x^2 - 3*y^2;
- x_val = subs(Tx, [x, y], [2, 1])
- y_val = subs(Ty, [x, y], [2, 1])

Slide 3 · Partial Derivative-Class II

- Bm_value = subs(Bm, {m, h}, {m_value, h_value});
- Bh_value = subs(Bh, {m, h}, {m_value, h_value});

Slide 4 · Partial Derivative-Class II

- 3) A balloon is in the form of right circular cylindrical of radius 1.5m and length 4m and is Surmounted by hemispherical end. If the radius is increased by 0.01 m and length by 0.05m, find the percentage change in volume of balloon.
- dV = diff(V, r) * dr + diff(V, h) * dh;
- dV_value = double(subs(dV, {r, h, dr, dh}, {r_val, h_val, dr_val, dh_val}));

Slide 5 · Partial Derivative-Class II

- 4) Divide 24 into 3 parts such that the continued product of the first, square of the second and the cube of the third may be maximum.
- sol = solve([eq1, eq2, eq3, eq4], [x, y, z, lambda], 'Real', true);
- if x_vals(i) > 0 && y_vals(i) > 0 && z_vals(i) > 0

Slide 6 · Partial Derivative-Class II

- 5) If x increases at the rate of 2cm/sec at the instant when x=3cm and y=1cm .At what rate must y be changing in the order that the function 2xy-3x^2y shall be neither increasing nor decreasing?
- syms x y dxdt dydt t;
- dydt_value = subs(dydt_expr, {x, y, dxdt}, {3, 1, dxdt_value})

Slide 7 · Partial Derivative-Class II

- % Define the objective function and constraint
- budget_constraint = 200*x + 100*y - 10000;
- solution = solve([dL_dx, dL_dy, dL_dlambda], [x, y, lambda]);

Slide 8 · Partial Derivative-Class II

- % Extract the solution (only real, positive values)
- % Filter out non-positive solutions (if any)
- fprintf('Maximum Q(x, y): %.2e\n', double(subs(Q, [x, y], [x_opt,y_opt])));

Slide 9 · Partial Derivative-Class II

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 10 · Partial Derivative-Class II

- % Define battery charge (%) and speed (km/h)
- [X,Y] = meshgrid(x,y);
- R = 3*X - 0.02*Y.^2 + 500;

Slide 11 · Partial Derivative-Class II

- % Partial derivatives using symbolic variables
- disp('Partial derivative with respect to battery charge:')
- disp('Partial derivative with respect to speed:')

Slide 12 · Partial Derivative-Class II

- fprintf('\nAt Battery Charge = 80%% and Speed = 100 km/h:\n');
- fprintf('dR/dx = %.2f km per %% battery charge\n',double(dR_dx_val));
- fprintf('dR/dy = %.2f km per km/h speed\n',double(dR_dy_val));

Slide 13 · Partial Derivative-Class II

- Partial derivative with respect to battery charge:

- Partial derivative with respect to speed :
- At BatteryCharge=80%and Speed=100km/h:

Slide 14 · [Image text, OCR]

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000676 · EV0000677 · EV0000678 · +39 source units in pack.json

11 Source slide deck: UNIT 2

Study focus: Source slide deck: UNIT 2. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000718 · EV0000719

SHORT-NOTE RECAP

Study notes · Source slide deck: UNIT 2

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · [Image text, OCR]

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 2 · Engineering MATHEMATICS - II

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · ENGINEERING MATHEMATICS-I

- To solve Ordinary differential equation,
- Symbolic Solution using dsolve command
- Numeric Solution using ode45 command

Slide 4 · ENGINEERING MATHEMATICS-I

- S = dsolve(eqn) solves the differential equation eqn, where eqn is a symbolic equation.
- Use diff and == to represent differential equations. For example, diff(y,x) == y
- represents the equation dy/dx = y. Solve a system of differential equations by specifying eqn as a vector of those equations.

Slide 5 · ENGINEERING MATHEMATICS-I

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 6 · Higher order differential equations:

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 7 · Higher order differential equations:

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 8 · Higher order differential equations, Continued...

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 9 · Higher order differential equations, Continued...

- (5 exp((3 x)/2))/2 - (3 exp(x/2))/2

Slide 10 · Higher order differential equations, Continued...

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 11 · Higher order differential equations, Continued...

- ySol =2*exp(2*x) - 3*exp(3*x) + exp(x)
- 2 exp(2 x) -3 exp(3 x) + exp(x)

Slide 12 · Non-Homogeneous differential equations

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 13 · Non-Homogeneous differential equations, Continued...

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 14 · Non-Homogeneous differential equations, Continued...

- Sol =exp(-2*x)*(2*sin(3*x) - 9*cos(3*x) + 6))/3
- (exp (-2x)(2sin(3x)-9cos(3x)+6))/3

Slide 15 · Non-Homogeneous differential equations, Continued...

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 16 · Non-Homogeneous differential equations, Continued...

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 17 · Non-Homogeneous differential equations, Continued...

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 18 · Non-Homogeneous differential equations, Continued...

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 19 · Non-Homogeneous differential equations, Continued...

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 20 · Higher order differential equations, Continued...

- ode=diff(y,x,3)+2*diff(y,x,2)+diff(y,x,1)== x^2*exp(2*x)+sin^2(x);

Slide 21 · Higher order differential equations:

- ((2*x+3)^2)*D2y
- y-2*(2*x+3)*Dy-12*y==

Slide 22 · Higher order differential equations:

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 23 · Higher order differential equations:

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 24 · ENGINEERING MATHEMATICS-I

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 25 · [Image text, OCR]

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000718 · EV0000719 · EV0000720 · +72 source units in pack.json

12 Source slide deck: UNIT 2 CONTINUED

Study focus: Source slide deck: UNIT 2 CONTINUED. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000793 · EV0000794

SHORT-NOTE RECAP

Study notes · Source slide deck: UNIT 2 CONTINUED

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · [Image text, OCR]

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 2 · Higher order differential equations:

- Assume an object weighing 2 lb stretches a spring 6 in. The spring is released from the equilibrium position with an upward velocity of 16 ft/sec. Form the differential equation governing this motion and solve it. Find the period of the motion?
- ans = -2 sin(8 t)
- dsolve(ode, x(0)==0,Dx(0)==-16)

Slide 3 · Higher order differential equations:

- 2. At t=0, a current of 2 amperes flows in an RLC circuit with resistance R=40 ohms, inductance L=0.2 henrys, and capacitance C=10−5 farads. Find the current flowing in the circuit at t>0, if the initial charge on the capacitor is 1 coulomb. Assume that E(t)=0 for t>0.
- Qsol(x) = dsolve(ode,Q(0)==1,DQ(0)==2)
- e100x(10200 cos(200 x) 2000sin(200 x) _e-100 (100 cos(200 x)+ 51 sin(200 x)

Slide 4 · Higher order differential equations:

- 3.A spring with a mass of 2 kg has natural length 0.5m. A force of 25.6N is required to maintain it stretched to a length of 0.7m. The spring is immersed in a fluid with damping constant c = 40. Find the position of the mass at any time if it is pushed from the equilibrium position with an initial velocity of 0.6 m/s.
- ode=D2x+20*Dx+64*x==0
- dsolve(ode,x(0)==0,Dx(0)==0.6)

Slide 5 · Higher order differential equations:

- 4. A series circuit contains a resistor with R=24ohms, an inductor with L=2H, a capacitor with C=0.005F, and a 12V battery. The initial charge is Q=0.001C and the initial current is 0.
- Find the charge and current at time t.
- Q=dso1ve(D2Q+12*DQ+100*Q==6, Q(0)==0.001,DQ(0)==0)

Slide 6 · Higher order differential equations:

- 5. Consider an RLC series circuit with a resistor of 0.06ohm and an inductor of 0.01H, and a capacitor of 50/89 F powered by a voltage source E(t) = 0.1sin(10t)V . Initially, the current and charge on the capacitor are zero. Determine the current in the circuit as a function of time.
- 600 ee(13)e b
- (D2I+6*DI+178*I==100*c0s(10*t), I(0)==0,DI(0)==0)

Slide 7 · Higher order differential equations:

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 8 · Higher order differential equations:

- D1(t)=diff(I,1);

- D2(t)=diff(I,2);
- f=0.1*D2(t)+11*D1(t)+100*I==100*377*c0s(377*t);

Slide 9 · Higher order differential equations:

- f=D2(t)+6*D1(t)+18*Q==180;
- qsol(t)=dsolve(f,Q(0)==0,D1(0)==9)
- qsol(t) = 10 - 7 sin(3 t)e-3t - 10 cos(3t)e-3t

Slide 10 · Higher order differential equations:

- D1(t)=diff(I,1);
- D2(t)=diff(I,2);
- f=D2(t)+40*D1(t)+4*I==0;

Slide 11 · Higher order differential equations:

- 9. A 2kg mass is attached to a spring with a spring constant of 8N/m. The mass is displaced 0.5m from its equilibrium position and released from rest. Model the motion using a differential equation and find the position of the mass as a function of time.
- D1(t)=diff(x,1);
- f=D2(t)+4*x==0;

Slide 12 · Higher order differential equations:

- 10. In a spring-mass system, a mass attached to a spring oscillates when displaced from its equilibrium position. Without damping, the system would oscillate indefinitely (simple harmonic motion). However, in reality, a damping force (e.g., viscous friction proportional to velocity) opposes the motion, reducing the amplitude of oscillations over time. The system eventually comes to rest at the equilibrium position.
- A car suspension system can be modelled as a spring-mass system with a 100 kg mass (quarter-car model), a spring constant of 40000 N/m, and a damping coefficient of 2000 Ns/m. The car hits a bump, displacing the suspension 0.1 m upward, with an initial velocity of 0 m/s. Model the motion and find the time it takes for the amplitude to reduce to 0.01 m due to damping.

Slide 13 · Higher order differential equations:

- D1(t)=diff(x,1);
- f=D2(t)+20*D1(t)+400*x==0;
- dsolve(f,x(0)==0.1,D1(0)==0)

Slide 14 · Higher order differential equations:

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 15 · Higher order differential equations:

- Let c(t) be the displacement of the machine from its equilibrium position.

- Using Newton's Second Law,
- F(t) = 300 cos(5t)

Slide 16 · Higher order differential equations:

- ode = diff(x,t,2) + 90*x == 2*cos(5*t);
- fplot(xSol,[0 10],'LineWidth',2)
- title('Displacement of Vibrating Press Machine')

Slide 17 · Higher order differential equations:

- 01 = 15 (6 √10 - 10)
- 03 = t (3 √10 - 5)
- 04 = t (3 √10+ 5)

Slide 18 · Higher order differential equations:

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 19 · Higher order differential equations:

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 20 · Higher order differential equations:

- Using Newton's Second Law,
- 1200x" + 96000c = 2400 cos(6t)

Slide 21 · Higher order differential equations:

- ode = diff(x,t,2) + 80*x == 2*cos(6*t);
- fplot(xSol,[0 20],'LineWidth',2)
- title('Response of Railway Coach Suspension')

Slide 22 · Higher order differential equations:

- 20 (2 5 -3) (2 √5 + 3)
- 01 =t (4 v5 - 6)
- 02 = t (4 √5 + 6)

Slide 23 · Higher order differential equations:

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 24 · [Image text, OCR]

- Department of Science and Humanities

EV0000793 · EV0000794 · EV0000795 · +69 source units in pack.json

13 Source slide deck: UNIT 3

Study focus: Source slide deck: UNIT 3. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000865 · EV0000866

SHORT-NOTE RECAP

Study notes · Source slide deck: UNIT 3

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · [Image text, OCR]

- Department of Science and Humanities

Slide 2 · Function | Purpose

- | Partial derivative of f w.r.t. x
- | Isolate a variable from equation
- | Replace a symbol with expression

Slide 3 · %% Step 1: Define symbolic variables

- %% Step 2: Define the surface equation
- %% Step 4: Solve for arbitrary constants a and b
- %% Step 5: Substitute back into z to eliminate a and b

Slide 4 · Form the PDE by eliminating arbitrary constants: z = (x + a) (y + b)

- %% Step 2: Define the surface equation
- %% Step 4: Solve for arbitrary constants a and b
- %% Step 5: Substitute back into z to eliminate a and b

Slide 5 · %% Step 1: Define symbolic variables

- %% Step 2: Define the surface equation
- %% Step 4: Solve for arbitrary constants a and b
- %% Step 5: Substitute back into z to eliminate a and b

Slide 6 · %% Step 1: Define symbolic variables

- %% Step 2: Define the surface equation
- %% Step 4: Solve for arbitrary constants a and b
- %% Step 5: Substitute back into z to eliminate a and b

Slide 7 · %% Step 1: Define symbolic variables

- %% Step 2: Define the surface equation
- %% Step 3: Compute partial derivatives
- %% Step 4: Solve for function f'(u)

Slide 8 · %% Step 1: Define symbolic variables

- %% Step 2: Define the surface equation
- %% Step 3: Compute partial derivatives
- %% Step 4: Solve for function f'(u)

Slide 9 · %% Step 1: Define symbolic variables

- %% Step 2: Define the surface equation
- %% Step 3: Compute first order partial derivatives
- %% Step 4: Compute second order partial derivatives

Slide 10 · Separation of Variables

- Symbolic eq object | PDE = diff(u, t)
- = = diff(u, x, 2) | Creates equation,
- substitution | Subs(PDE, u, X(x)*T(t)) | Replaces u with X(x)*T(t)

Slide 11 · %% Step 1: Define symbolic variables

- %% Step 2: Define the PDE
- %% Step 3: Using separation of Variables
- %% Step 4: Substituting u(x,t) and dividing the PDE with u(x,t)

Slide 12 · Write code for symbolic solution of the 1D Heat Equation Using Separation of Variables

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 13 · Write code for solution of PDE using Separation of Variables

- %% Step 1: Define symbolic variables
- %% Step 3: Using separation of Variables
- %% Step 4: Substituting u(x,y) and dividing the PDE with u(x,y)

Slide 14 · Write code for solution of PDE using Separation of Variables

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 15 · Write code for solution of PDE using Separation of Variables

- %% Step 1: Define symbolic variables
- %% Step 3: Using separation of Variables
- %% Step 4: Substituting u(x,t) and dividing the PDE with u(x,t)

Slide 16 · Write code for solution of PDE using Separation of Variables

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 17 · %% Step 1: Define symbolic variables

- %% Step 2: Define the PDE
- %% Step 3: Using separation of Variables
- %% Step 4: Substituting u(x,y) and dividing the PDE with u(x,y)

Slide 18 · Slide page 18

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 19 · Partial Differential Equations

- syms x y C1 C2
- z = C1*exp(2*y+x) + C2*exp(3*y+x);
- f = matlabFunction(z,'Vars',[x y C1 C2]);

Slide 20 · Partial Differential Equations

- Cex+2y + C2 ex+3 y

Slide 21 · Partial Differential Equations

- MATLAB cannot automatically compute the Particular Integral (PI) using the operator method

Slide 22 · [Image text, OCR]

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000865 · EV0000866 · EV0000867 · +63 source units in pack.json

14 Source slide deck: UNIT 3 CONTINUED

Study focus: Source slide deck: UNIT 3 CONTINUED. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000931 · EV0000932

SHORT-NOTE RECAP

Study notes · Source slide deck: UNIT 3 CONTINUED

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · [Image text, OCR]

- Department of Science and Humanities

Slide 2 · Applications of Partial Differential Equation:

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · One-Dimensional Heat Equation

- The heat equation is a partial differential equation that models the distribution and flow of heat (or temperature) in a given region over time.
- Ione-dimensional heat equation helps to find distribution of heat in metal rod, cooling of electronic devices, heat flow in buildings, etc,.

Slide 4 · One-Dimensional Heat Equation

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 5 · One-Dimensional Heat Equation

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 6 · MATLAB code for the 1-D heat equation

- syms X(x) T(t)
- PDE = diff(u,t) == diff(u,x,2);
- PDE_sep = simplify(PDE/(X(x)*T(t)));

Slide 7 · MATLAB code for the 1-D heat equation

- eqX = diff(X(x), x, 2) == - lambda*X(x);
- % Final solution U_sol= simplify(X_sol*T_sol);
- T(t) = C1*exp(-lambda*t) X(x) = C1*cos(lambda^(1/2)*x) - C2*sin(lambda^(1/2)*x) U(x, t) = C1*exp(-lambda*t)*(C1*cos(lambda^(1/2)*x) - C2*sin(lambda^(1/2)*x))

Slide 8 · One-Dimensional Heat equation

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 9 · One-Dimensional Heat equation

- PDE = diff(u,t) == 4* diff(u,x,2);
- PDE_sep = simplify(PDE/(X(x)*T(t)));
- eqT = diff(T(t), t) + lambda*T(t) == 0;

Slide 10 · One-Dimensional Heat equation

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 11 · One-Dimensional Heat equation

- PDE = diff(u,t) == 16* diff(u,x,2);
- PDE_sep = simplify(PDE/(X(x)*T(t)));

- f=100*cos(pi*x/2);

Slide 12 · One-Dimensional Heat equation

- eqX = 16*diff(X(x), x, 2) + lambda*X(x) == 0;
- T(t) = 100*exp(-lambda*t)*cos((pi*x)/2)
- U(x,t)=(100*sin((lambda^(1/2)*x)/4)

Slide 13 · One-Dimensional Heat equation

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 14 · One-Dimensional Heat equation

- syms u(x, t) x t % Symbolic variables and functions
- u(x, t) = X(x) * T(t); % Assume u(x,t) = X(x)T(t)
- T_sol = dsolve(eqT, T(0) == 100*x*(1-x))% Arbitrary initial condition for T

Slide 15 · One-Dimensional Heat equation

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 16 · [Image text, OCR]

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000931 · EV0000932 · EV0000933 · +45 source units in pack.json

15 Source slide deck: UNIT 4

Study focus: Source slide deck: UNIT 4. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000979 · EV0000980

SHORT-NOTE RECAP

Study notes · Source slide deck: UNIT 4

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · [Image text, OCR]

- Department of Science and Humanities

Slide 2 · Theory and Visualization

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · Theory and Visualization

- %% Visualizing Gamma (2D) and Beta (3D Surface)
- x_gamma = linspace(0.1, 5, 200);
- plot(intervals, gamma(intervals), 'ro', 'MarkerFaceColor', 'r', 'MarkerSize', 6);

Slide 4 · Theory and Visualization

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 5 · Evaluation of Gamma function

- 5. gamma(1/4)*gamma(3/4)

Slide 6 · Evaluation of Gamma function

- 6. Evaluate several values of the gamma function between [-3.5, 3.5].
- x = -3.5:3.5;
- 0.2701 -0.9453 2.3633 -3.5449 1.7725 0.8862 1.3293 3.3234

Slide 7 · Evaluation of Gamma function

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 8 · Gamma function

- Geometrical representation of Gamma function:

Slide 9 · Evaluation of Beta function

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 10 · Evaluation of Beta function from Gamma function

- %Relation between gamma and beta functions
- y2 = (gamma(2) * gamma(3)) / gamma(5)

Slide 11 · p=10;

- result=(gamma((p+1)/2)*gamma(1/2))/(2*gamma((p+2)/2));

Slide 12 · m = 7;

- result = (gamma(m/2) * gamma(n/2)) / (2 * gamma(m/2 + n/2))

Slide 13 · p= 1/2;

- result = (1/3) * (gamma(p) * gamma(q)) / gamma(p + q)

Slide 14 · Problem 1

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 15 · Problem 1

- % Define the integrand f = @(x) x.^(5/2).*exp(-x); % Compute mass using Gamma function M_gamma = gamma(7/2);
- f = x^(5/2)*exp(-x)
- % Area using MATLAB beta function

Slide 16 · Problem 2

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 17 · Problem 2

- % Define the integrand f = @(x) x.^(1/2).*(1-x).^(3/2); % Area using MATLAB beta function A_beta = beta(3/2,5/2);
- f = x^(1/2)*(1-x)^(3/2)
- A_beta = beta(3/2,5/2)

Slide 18 · [Image text, OCR]

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000979 · EV0000980 · EV0000981 · +51 source units in pack.json

16 Source slide deck: UNIT 4 CONTINUED

Study focus: Source slide deck: UNIT 4 CONTINUED. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0001033 · EV0001034

SHORT-NOTE RECAP

Study notes · Source slide deck: UNIT 4 CONTINUED

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · [Image text, OCR]

- Department of Science and Humanities

Slide 2 · Domain-specific Engineering application

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · Domain-specific Engineering application

- % Solve using the built-in gamma function
- % Since Gamma(2) = 1! = 1, the mean should be 4 (4,000 hours)

- mean_lifespan = (2^2) * gamma(2)

Slide 4 · Domain-specific Engineering application

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 5 · Domain-specific Engineering application

- %Simple Digital Switch Active-Time Profile
- x = linspace(0, 1, 100); % Active ratio from 0 to 1
- f_x = (x.^(a-1) .* (1-x).^(b-1)) / beta_val; % Evaluates to f(x)

Slide 6 · Domain-specific Engineering application

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 7 · Domain-specific Engineering application

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 8 · Domain-specific Engineering application

- % CS: 2-Package Software Compilation Delay
- t = linspace(0, 8, 200); % Time scale from 0 to 8 seconds
- f_t = (t .* exp(-t)) / gamma_val; % Evaluates to t*e^(-t)

Slide 9 · Domain-specific Engineering application

- Total Time (t) in Seconds

Slide 10 · Domain-specific Engineering application

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 11 · Domain-specific Engineering application

- time = 0:0.1:8; % Time in seconds
- k = 2; theta = 1; % 2 parts, 1 sec each
- % Using the gamma function to draw the prediction curve

Slide 12 · Domain-specific Engineering application

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 13 · Domain-specific Engineering application

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 14 · Domain-specific Engineering application

- x= linspace(0, 1, 100);

- % Using the beta function to draw the likelihood curve
- xlabel('Click Percentage (0 to 1)'); ylabel('Likelihood');

Slide 15 · Domain-specific Engineering application

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 16 · [Image text, OCR]

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0001033 · EV0001034 · EV0001035 · +45 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Introduction to MATLAB Software transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/001_Introduction_to_MATLAB_Software__60472b361e5bb782.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to MATLAB Software transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/001_Introduction_to_MATLAB_Software__60472b361e5bb782.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to MATLAB Software transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/001_Introduction_to_MATLAB_Software__60472b361e5bb782.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Introduction to MATLAB Software transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/001_Introduction_to_MATLAB_Software__60472b361e5bb782.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Data types; relational operators; logical operators transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/002_Data_types_relational_operators_logical_operators__915d8f716427b954.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Data types; relational operators; logical operators transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/002_Data_types_relational_operators_logical_operators__915d8f716427b954.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Data types; relational operators; logical operators transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/002_Data_types_relational_operators_logical_operators__915d8f716427b954.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.
Data types; relational operators; logical operators transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/002_Data_types_relational_operators_logical_operators__915d8f716427b954.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
M-Files; Loop Control Statements; Conditional Statements transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/003_M-Files_Loop_Control_Statements_Conditional_Statements__d8fc5a0a5b8f6dbe.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
M-Files; Loop Control Statements; Conditional Statements transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/003_M-Files_Loop_Control_Statements_Conditional_Statements__d8fc5a0a5b8f6dbe.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
M-Files; Loop Control Statements; Conditional Statements transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/003_M-Files_Loop_Control_Statements_Conditional_Statements__d8fc5a0a5b8f6dbe.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9986; only the JSON segmentation supplies the displayed timestamps.
M-Files; Loop Control Statements; Conditional Statements transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/003_M-Files_Loop_Control_Statements_Conditional_Statements__d8fc5a0a5b8f6dbe.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/004_Plotting_of_Curves_2D_and_3D_Plot_Labeling_Curve_Labeling__e96c8141e2597d5f.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/004_Plotting_of_Curves_2D_and_3D_Plot_Labeling_Curve_Labeling__e96c8141e2597d5f.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/004_Plotting_of_Curves_2D_and_3D_Plot_Labeling_Curve_Labeling__e96c8141e2597d5f.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/004_Plotting_of_Curves_2D_and_3D_Plot_Labeling_Curve_Labeling__e96c8141e2597d5f.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Video transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/006_Video__bb969ddbec162496.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Video transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/006_Video__bb969ddbec162496.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Video transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/006_Video__bb969ddbec162496.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9998; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Video transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/006_Video__bb969ddbec162496.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Video transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/008_Video__c789c482b6a9141a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Video transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/008_Video__c789c482b6a9141a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Video transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/008_Video__c789c482b6a9141a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9997; only the JSON segmentation supplies the displayed timestamps.
Video transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/008_Video__c789c482b6a9141a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Slides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Slides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
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Slide page text: Slides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Slides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Slides (converted PDF; original hash 6ae797d02b605192a59363cd34046d94d555b15ec6da51461b7dc3575b6bf74f)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Slides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Slides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Slides (converted PDF; original hash 9988a37be0a34576dd47bea022ec4b865fabb5177da0c2acbb6534e78056d068)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Slides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Slides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Slides (converted PDF; original hash 2e5770ab70ab393896a1c030e9188dd0b9bff0b94ee6896928faa6ebc98aaf4e)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: MATLAB INTRODUCTION	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: MATLAB INTRODUCTION	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for MATLAB INTRODUCTION (converted PDF; original hash e6958cd9f18100fdd7d6aeaa16622f9e9dc13fd36b1538aeb854e3b5f44de850)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UNIT 1	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UNIT 1	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UNIT 1 (converted PDF; original hash b9b09a103cf726f6b5ce82506486e7950cc58cc08192d2b0282d209386130451)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UNIT 1 CONTINUED	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UNIT 1 CONTINUED	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UNIT 1 CONTINUED (converted PDF; original hash cbb14f9556b8abbb9d56b7ef343a57527b2bc4fdc9129f71b84d71b9a2af6bff)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UNIT 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UNIT 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UNIT 2 (converted PDF; original hash 125940f7b2e0b5146087b4d7dac7a42562e2a5ac72fcf50b71e36f58a1c3335f)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UNIT 2 CONTINUED	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UNIT 2 CONTINUED	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UNIT 2 CONTINUED (converted PDF; original hash cc656367591e70ecee8c8c2bed5168c4be97bb5375bb0cd112c313c359a66be8)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UNIT 3	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UNIT 3	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UNIT 3 (converted PDF; original hash 878ecae3697008466dc626559d5d483e7430220a9c80ff9b52d3d11c0f58d20c)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UNIT 3 CONTINUED	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UNIT 3 CONTINUED	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UNIT 3 CONTINUED (converted PDF; original hash 9700aa5026207dd1afba49fa12f067da4dd78a34edac2c8722499d9e00cf13c6)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UNIT 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UNIT 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UNIT 4 (converted PDF; original hash 3b995ad93f2cb3ec614459bbc2a4ac4770165b8c527997a60e8257c0bb457442)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UNIT 4 CONTINUED	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UNIT 4 CONTINUED	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UNIT 4 CONTINUED (converted PDF; original hash 1d2300b5e26a0738a13cd90afdb8aece95f5fb6bab9de84ca9db21e34a148039)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 4 rendered pages in 001_Slides__7ac00f797de85ee4.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 002_Slides__ef919841425356a4.slides.md have no readable `.slides.md` text; their original page images are included.
3. 1 rendered pages in 003_Slides__3d7af4a88d5b4490.slides.md have no readable `.slides.md` text; their original page images are included.
4. 1 rendered pages in 004_Slides__4150cd4c33be9a2c.slides.md have no readable `.slides.md` text; their original page images are included.
5. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
6. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
7. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
8. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
9. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
10. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
11. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00001	1 covered · 0 excluded · 0 gaps	catalogue: Unit title and index copied from the official units.json catalogue; human content review is pending.
SRC00002	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0060. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0061 to s0119. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0120 to s0176. Sidecar variants compare...
SRC00003	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0060. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0061 to s0119. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0120 to s0176. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00004	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0060. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0061 to s0119. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0120 to s0176. Sidecar variants compare...
SRC00005	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0060. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0061 to s0119. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0120 to s0176. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00006	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...
SRC00007	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00008	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...
SRC00009	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00010	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0035 to s0051. Sidecar variants compare...
SRC00011	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0035 to s0051. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00012	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0035 to s0051. Sidecar variants compare...
SRC00013	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0035 to s0051. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00014	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...
SRC00015	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00016	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...
SRC00017	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00018	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0063. Sidecar variants compare...
SRC00019	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00020	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0063. Sidecar variants compare...
SRC00021	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00022	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0055. Sidecar variants compare...
SRC00023	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0055. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00024	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0055. Sidecar variants compare...
SRC00025	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0055. Sidecar variants compare...
SRC00026	0 covered · 0 excluded · 41 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...

Source	Unit counts	Example decisions
SRC00028	0 covered · 0 excluded · 45 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00027	0 covered · 0 excluded · 41 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00029	0 covered · 0 excluded · 25 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00031	0 covered · 0 excluded · 27 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00030	0 covered · 0 excluded · 25 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00032	0 covered · 0 excluded · 23 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00034	0 covered · 0 excluded · 24 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00033	0 covered · 0 excluded · 23 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00035	29 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00037	30 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00036	29 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00038	0 covered · 0 excluded · 30 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00040	0 covered · 0 excluded · 30 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00039	0 covered · 0 excluded · 30 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00041	38 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00043	38 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00042	38 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00044	14 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00046	14 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00045	14 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00047	25 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00049	25 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00048	25 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00050	24 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00052	24 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00051	24 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00053	22 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00055	22 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00054	22 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00056	16 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00058	16 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00057	16 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00059	18 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00061	18 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00060	18 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00062	16 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00064	16 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00063	16 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

C Revision and retrieval practice